TLS Fingerprinting: Phase 2 chat

Su and Claude · 2 to 6 November 2026 · CS331 Computer Networks, Group 13

SU

Hello, so obviously yk this right?
https://github.com/milonithakkar/CS331-Phase2
https://docs.google.com/document/d/19G069v5OEzf9qigTIG8ZSF2_GKOpRCowl-NK72--e-M/edit

CLAUDE

Yes, I know this one. It's your Group 13 Phase 2 plan: you take the existing fingerprinting tools (FoxIO JA4+, Wireshark/tshark, and ja4db.com for lookups) and find where they break in practice.

The doc has grown since the short objective.md you sent Sameer sir. It now covers:

- Main goal: map the current landscape, then find 6–7 practical limitations, each backed by packet captures, with an explanation of what changes at the protocol level and which RFC defines it. Then go deep on 1–2 of them and end with a fix, or a clear account of what a fix would need.
- Basics: for each fingerprint, document its fields and their layers, the RFC behind each field, how the hash is built (MD5 for JA3, truncated SHA-256 sections for JA4), and that it identifies an app, not a user.
- 9 scenarios, changing one thing at a time: different OS, app versions, repeat connections, apps sharing a TLS library, IPv4 vs IPv6, TCP vs QUIC, post-quantum on/off, ECH on/off, plain vs encrypted DNS.
- What you measure: whether the tool parses at all, whether the hash is stable, whether ja4db matches the right app, and how often different apps collide.
- Plan: compare your Project 1 tool against FoxIO and Wireshark, establish a baseline, run the scenarios, pick the failures to study, and try one fix.

I couldn't open the GitHub repo. milonithakkar/CS331-Phase2 gives a 404 from here, so it's either private or the name is slightly off. If Miloni adds you as a collaborator, or you paste the files, I can look at what's there.

What do you want to do next: start the basics section, set up the captures, or something else?

SU

added the folder, this is theimplementation folder.
Thing is, I need to learn (not the implementation, that is not that important) thw whole what why how and everything (theory) of things

CLAUDE

(Asked two questions first. Su picked: build it module by module, aiming for deep understanding.)

I've made Module 1, "Reading the ClientHello". It's 25 cards, and on each one you answer first and get the explanation after.

It covers:

- Why only the ClientHello and ServerHello can be read on the wire, and why clients produce different bytes: it comes down to the TLS library plus the app's settings.
- Walking the real bytes yourself, including legacy_version and GREASE.
- How JA3 is built, and why Chrome's extension shuffle breaks it.
- How each part of JA4 is built, and what a match does and doesn't tell you.
- JA3S and JA4S at the end.

The example hashes are real values, computed by running both recipes on one sample client.

Modules 2–6 follow your seven limitations: what TLS 1.3 hides, TCP and capture position, QUIC, post-quantum, and ECH with encrypted DNS. Tell me when you've finished Module 1 and which cards felt shaky, and I'll build Module 2 around those.

Lesson link: https://claude.ai/artifact/5r4MAzHVasGvKJCGtNC5CH

SU

ahh, why are u asking questions before even teaching?

CLAUDE

You're right, it was quizzing you on things it hadn't taught yet. I've fixed it at the same link: every card now opens with a short lesson, and you press "Check yourself" for a question that only uses what the lesson just covered.

I also cut the card count from 25 to 22. Two cards are merged into the lessons around them, and a few questions are reworded so they test the idea rather than the lesson's wording. While answering a question, you can tap "Re-read the lesson" to see it again.

I tested it in a browser and it loads and moves between cards without errors. Your old progress was reset, so you start from card 1.

SU

make a pdf of this chat of ours? will you? my questions + ur responses